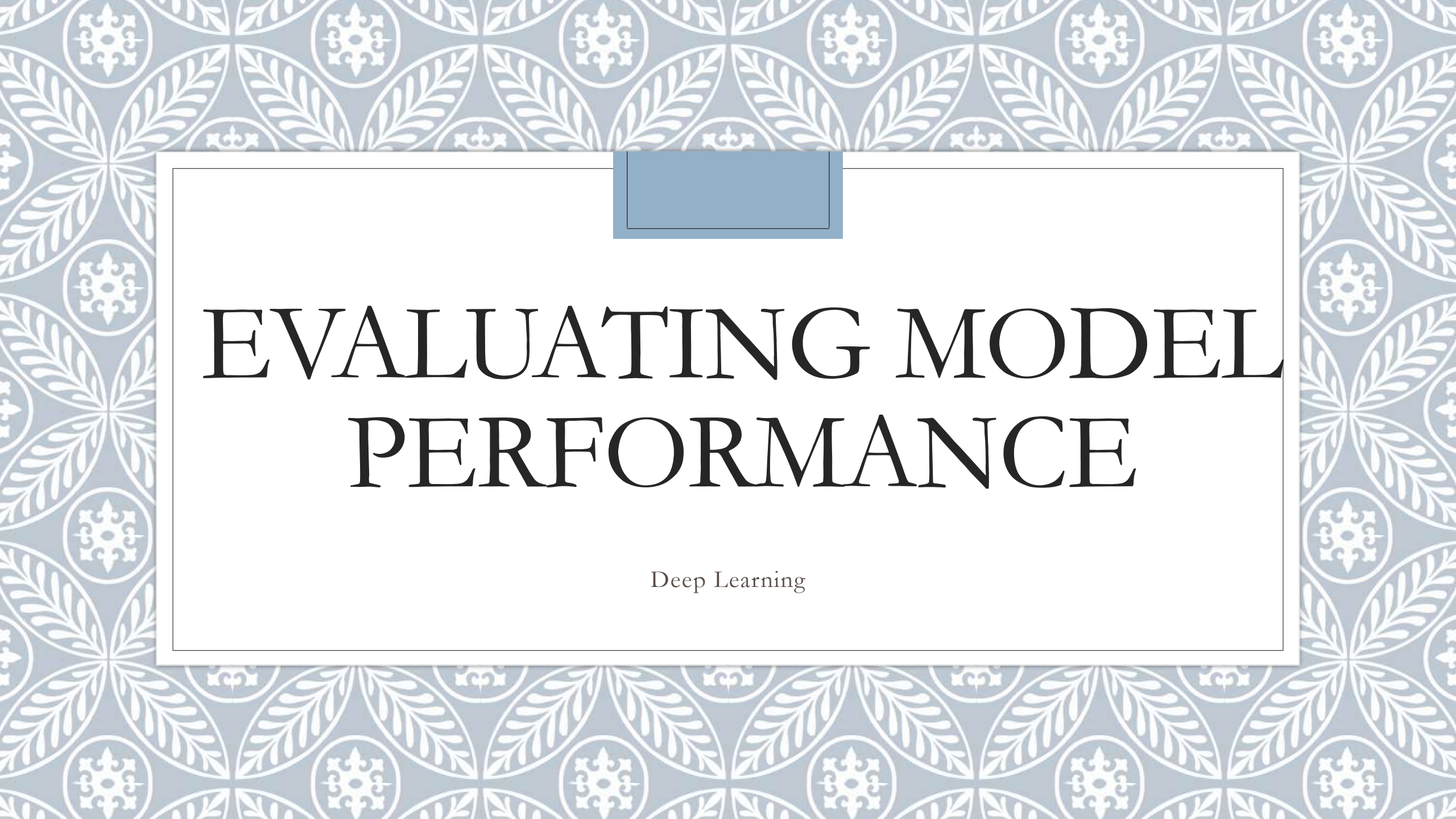




EVALUATING MODEL PERFORMANCE

Deep Learning

Evaluating model performance

- Precision
- F1 Score
- Recall
- True Positive

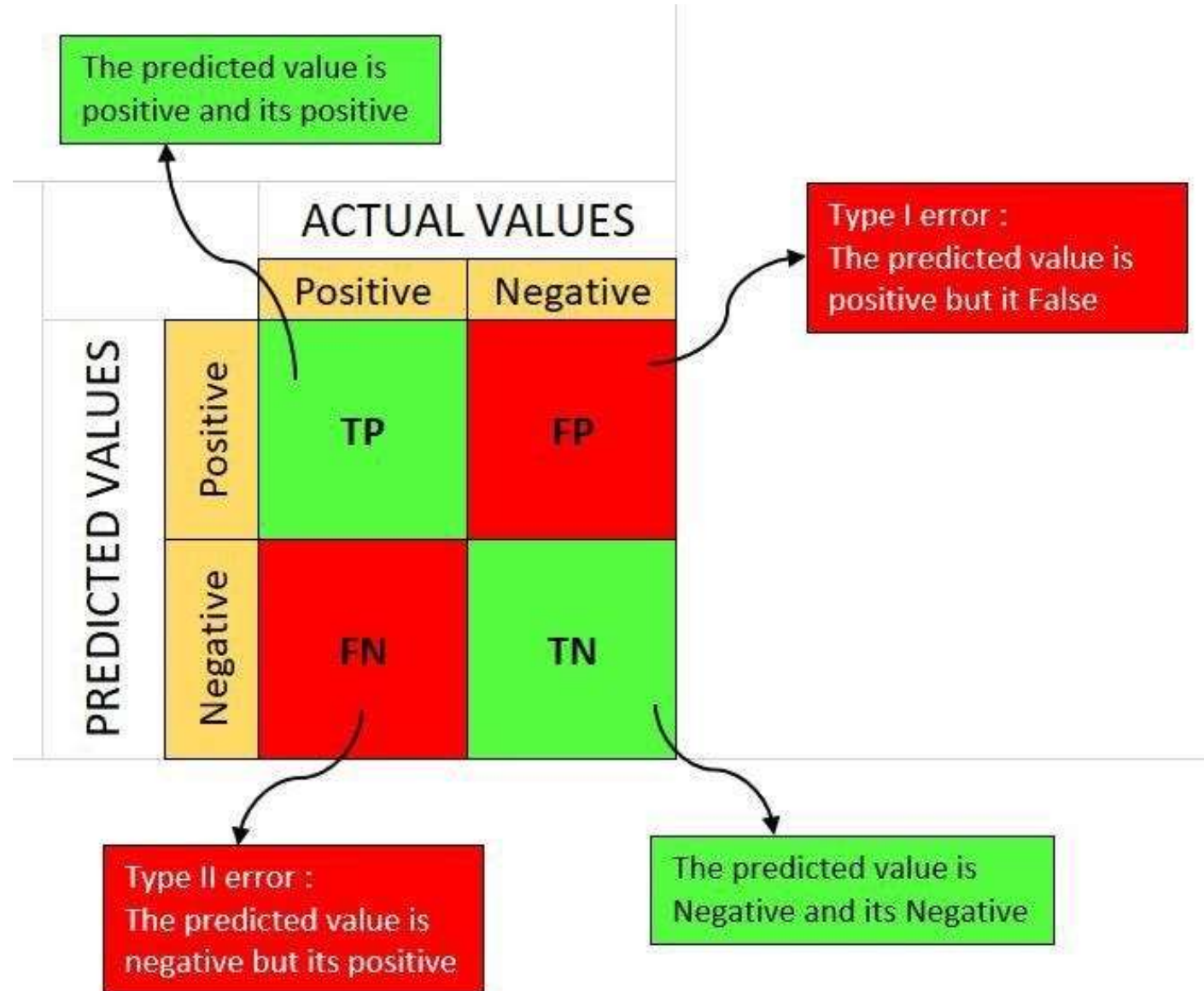
Confusion Matrix

- It is a performance measurement for machine learning classification problem where output can be two or more classes.
- It is a table with 4 different combinations of predicted and actual values.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Confusion Matrix

- **True Positive (TP):** The model correctly predicted a positive outcome (the actual outcome was positive).
- **True Negative (TN):** The model correctly predicted a negative outcome (the actual outcome was negative).
- **False Positive (FP):** The model incorrectly predicted a positive outcome (the actual outcome was negative). Also known as a Type I error.
- **False Negative (FN):** The model incorrectly predicted a negative outcome (the actual outcome was positive). Also known as a Type II error.



Metrics based on Confusion Matrix Data

- **Accuracy:**

- It is defined as the fraction of the number of examples that are correctly classified by the classifier to the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

- **Precision:**

- Precision measures the proportion of true positive predictions among all positive predictions made by the model. It is calculated as the ratio of true positives to the sum of true positives and false positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Metrics based on Confusion Matrix Data

- **Recall (Sensitivity):**

- Recall measures the proportion of true positive predictions among all actual positive instances in the dataset. It is calculated as the ratio of true positives to the sum of true positives and false negatives.

$$Recall = \frac{TP}{TP + FN}$$

- **F1-score:**

- F1-score is the harmonic mean of precision and recall. It provides a single metric that combines both precision and recall, balancing the trade-off between the two. It is calculated as:

$$F_1Score = \frac{2 * Recall . Precision}{Recall + Precision} = \frac{2TP}{2TP + FP + FN}$$

Analysis with Performance Measurement Metrics

- Based on the various performance metrics, we can characterize a classifier.
- We do it in terms of TPR, FPR, Precision and Recall and Accuracy
- **Case 1: Perfect Classifier**

When every instance is **correctly** classified, it is called the **perfect classifier**. In this case, $TP = P$, $TN = N$ and CM is

$$TPR = TP / (TP + FN) = \frac{P}{P} = 1$$

$$FPR = FP / (FP + TN) = \frac{0}{N} = 0$$

$$Precision = \frac{P}{P} = 1$$

$$F_1 \text{ Score} = \frac{2 \times 1}{1 + 1} = 1$$

$$Accuracy = \frac{P + N}{P + N} = 1$$

		Predicted Class	
		+	-
Actual class	+	P	0
	-	0	N

Analysis with Performance Measurement Metrics

◦ Case 1: Worst Classifier

When every instance is **wrongly** classified, it is called the **worst classifier**. In this case, $TP = 0$, $TN = 0$ and CM is

$$TPR = \frac{0}{P} = 0$$

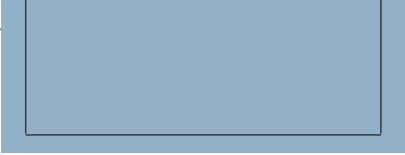
$$FPR = \frac{N}{N} = 1$$

$$Precision = \frac{0}{N} = 0$$

F_1 Score = Not applicable
as $Recall + Precision = 0$

$$Accuracy = \frac{0}{P+N} = 0$$

		Predicted Class	
		+	-
Actual class	+	0	P
	-	N	0



THANK YOU!!